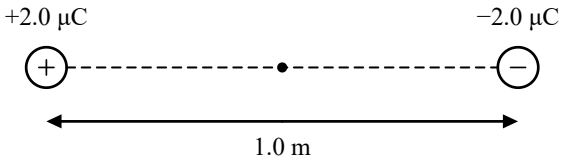
STPM 2024 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. Two point charges, +2.0 μC and −2.0 μC, are separated at a distance of 1.0 m is shown in the diagram below.



What is the magnitude and direction of the resultant electric field strength at the mid-point between the charges?

A 7.2×10^4 N C^{-1} towards the right
B 7.2×10^4 N C^{-1} towards the left
C 1.4×10^5 N C^{-1} towards the right
D 1.4×10^5 N C^{-1} towards the left

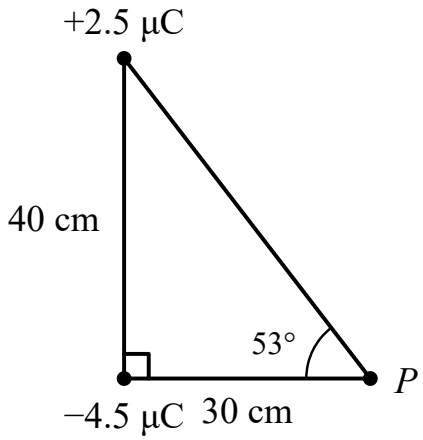
Answer: C

Each charge: $E = \frac{kq}{r^2} = \frac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{(0.50)^2} = 7.2 \times 10^4$ N C^{-1}

Both fields point to the right (away from +, towards −): total 1.4×10^5 N C^{-1} to the right.

Explanation: Between opposite charges the two fields add.

2. Two point charges, +2.5 μC and −4.5 μC, are separated at a distance 40 cm is shown in the diagram below.



What is the electric potential at *P*?

A -1.8×10^5 V B -9.0×10^4 V C $+9.0 \times 10^4$ V D $+1.8 \times 10^5$ V

Answer: B

Distance of P from +2.5 μC = $\sqrt{(40^2 + 30^2)}$ = 50 cm; from −4.5 μC = 30 cm.

$$V = (9.0 \times 10^9)\left(\frac{2.5 \times 10^{-6}}{0.50} - \frac{4.5 \times 10^{-6}}{0.30}\right) = -9.0 \times 10^4 \text{ V}$$

Explanation: Potential is a scalar: add the contributions with their signs.

3. A phone battery with energy storage of 19.2 kJ has the same amount of energy as a capacitor which is fully charged to 6.00 V. The circuit that connects the capacitor has an equivalent resistance of 10.0 Ω and the phone still functions when the voltage of the capacitor drops from 6.00 V to 2.50 V. What is the capacitance and duration of usage for the phone?

	Capacitance	*Duration*
A	530 F	2.88×10^3 s
B	530 F	4.67×10^3 s
C	1070 F	5.76×10^3 s
D	1070 F	9.37×10^3 s

Answer: D

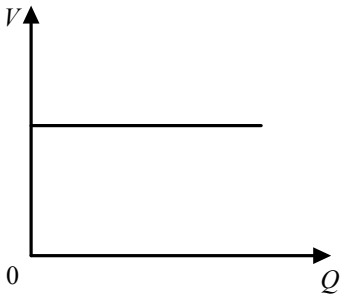
$$C = \frac{2U}{V^2} = \frac{2(19.2 \times 10^3)}{6.00^2} = 1.07 \times 10^3 \text{ F}$$

$$V = V_0 e^{-t/RC} \Longrightarrow t = RC \ln \frac{6.00}{2.50} = (10.0)(1070)(0.875) = 9.37 \times 10^3 \text{ s}$$

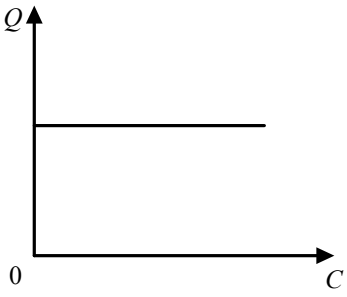
Explanation: The phone works while the capacitor discharges from 6.00 V to 2.50 V.

4. A capacitor of capacitance C is charged to charge Q so that the potential difference across the capacitor is V. Which graph shows an area under the graph that represents the energy stored by the capacitor?

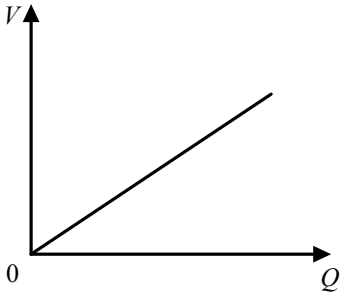
A



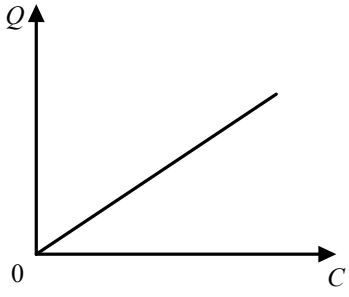
B



C



D



Answer: C

The energy stored is the area under a graph of V against Q: $\frac{1}{2}QV$ (C).

Explanation: Work done adding charge dQ at p.d. V is V dQ.

5. A bulb lights up almost immediately when the switch is closed although the drift velocity of the electrons in the circuit is low. Which is true about the situation?

A The resistance of the wire is very low.
B The density of free electrons is constant.
C The random velocity of the electrons is high.
D All free electrons drift simultaneously in the circuit.

Answer: D

The wires are already full of free electrons; when the switch is closed the electric field is set up almost instantly, so all free electrons start drifting at the same time.

Explanation: The signal travels at nearly the speed of light even though each electron drifts slowly.

6. The resistivity of copper is 1.7×10^{-8} Ω m. If the number of free electrons per unit volume in copper is 8.50×10^{28} m^{-3}, what is the average time between collisions of the electrons in a current-carrying copper wire?

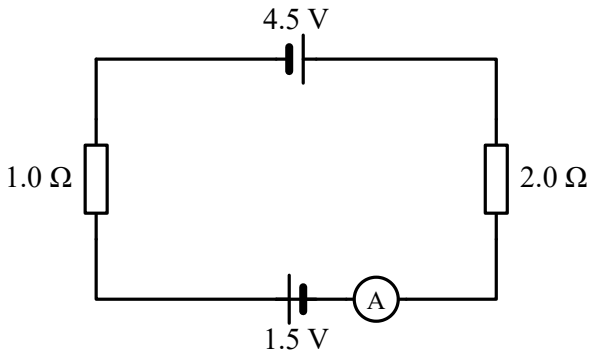
A 1.1×10^{-48} s B 3.9×10^{-31} s C 7.1×10^{-30} s D 2.5×10^{-14} s

Answer: D

$$\rho = \frac{m}{ne^2\tau} \Rightarrow \tau = \frac{9.11 \times 10^{-31}}{(8.50 \times 10^{28})(1.60 \times 10^{-19})^2(1.7 \times 10^{-8})} = 2.5 \times 10^{-14} \text{ s}$$

Explanation: The mean free time comes from the microscopic theory of conduction.

7. Two resistors that are connected to two batteries and ammeter are shown in the circuit diagram below.



What is the reading of the ammeter?

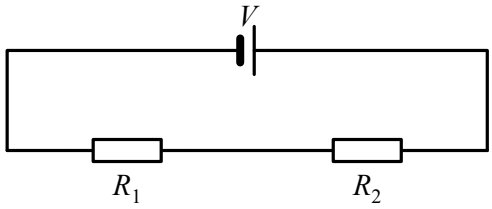
A 0.75 A B 1.0 A C 1.5 A D 2.0 A

Answer: D

The batteries aid each other: $I = \dfrac{4.5 + 1.5}{1.0 + 2.0} = 2.0\ \text{A}$

Explanation: In a single loop, add e.m.f.s that drive current the same way.

8. Two resistors of resistances R_1 and R_2 are connected in series to a battery of voltage V as shown in the circuit diagram below.



The potential difference across R_1 is

A $\dfrac{R_1}{R_2}V$ B $\dfrac{R_2}{R_1}V$ C $\dfrac{R_1}{R_2 + R_1}V$ D $\dfrac{R_2}{R_2 + R_1}V$

Answer: C

$$I = \frac{V}{R_1 + R_2} \Longrightarrow V_1 = IR_1 = \frac{R_1}{R_2 + R_1}V$$

Explanation: This is the potential-divider rule.

9. Which statement is **not** true about the motion of a charged particle that enters a uniform magnetic field?

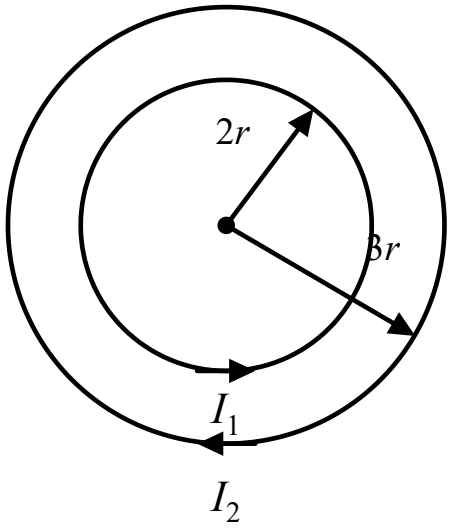
A The magnitude of the magnetic force is directly proportional to the charge of the particle.
B The path of the particle is parabolic if the direction of motion of the particle is parallel to the direction of the magnetic field.
C The path of the particle is circular if the direction of motion of the particle is perpendicular to the direction of the magnetic field.
D The magnitude of the magnetic force is maximum if the direction of motion of the particle is perpendicular to the direction of the magnetic field.

Answer: B

A particle moving parallel to the field experiences no magnetic force, so it moves in a straight line, not a parabola. Statement B is not true.

Explanation: $F = qvB \sin\theta$ is zero when $\theta = 0$.

10. Two concentric coils of radii $2r$ and $3r$, and currents I_1 and I_2, respectively are shown in the diagram below.



If the magnetic field strength at the centre of the coils is zero, what is the value of $\frac{I_1}{I_2}$?

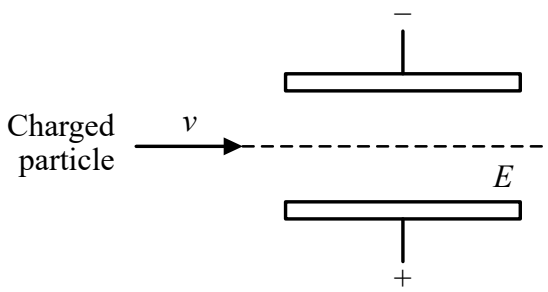
A $\frac{1}{3}$ B $\frac{2}{3}$ C $\frac{3}{2}$ D 2

Answer: B

$$\frac{\mu_0 I_1}{2(2r)} = \frac{\mu_0 I_2}{2(3r)} \Rightarrow \frac{I_1}{I_2} = \frac{2}{3}$$

Explanation: The currents flow in opposite directions so their fields at the centre cancel.

11. A positive charged particle enters perpendicularly to a uniform electric field, E, with a velocity, v, is shown in the diagram below.



Uniform magnetic field is applied to maintain the path of the charged particle in a straight line. What is the magnitude and direction of the magnetic field?

	Magnitude	*Direction*
A	$\frac{E}{v}$	Into the page
B	$\frac{E}{v}$	Out of page
C	$\frac{v}{E}$	Into the page
D	$\frac{v}{E}$	Out of page

Answer: B

The electric force on the positive particle is upward (from + plate towards − plate); the magnetic force must be downward.

$$qE = qvB \Longrightarrow B = \frac{E}{v}, \text{ directed out of the page}$$

Explanation: Use Fleming's left-hand rule (or $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$) to find the field direction.

12. When power is supplied to an electric motor, the motor armature rotates in a magnetic field and an induced e.m.f. is generated. Which statement is **not** true about the motor?

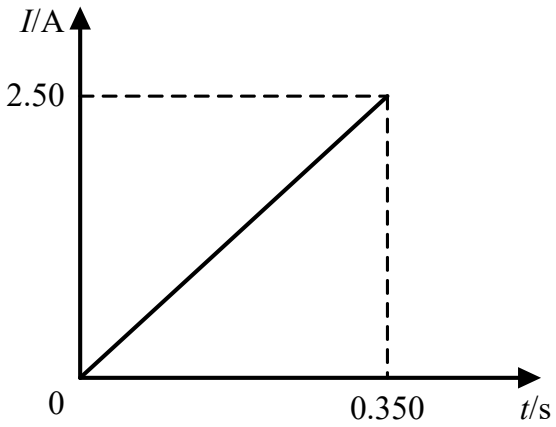
A The induced e.m.f. is maximum when the armature rotates with a maximum speed.
B The different voltage across the armature is smaller than that of the e.m.f. supplied.
C The induced e.m.f. is greater than the e.m.f. supplied.
D The induced e.m.f. opposes the e.m.f. supplied.

Answer: C

The back e.m.f. is always less than the supply e.m.f. (otherwise no current would flow). Statement C is not true.

Explanation: The back e.m.f. opposes the supply and is largest at full speed.

13. The variation of applied current, I, with time, t, for 200 turns solenoid of length 8.00 cm is shown in the graph below.



If the induced e.m.f. is 0.0150 V, what is the cross-sectional area of the solenoid?

A 2.10×10^{-3} m^2 B 3.34×10^{-3} m^2 C 5.22×10^{-1} m^2 D 6.68×10^{-1} m^2

Answer: B

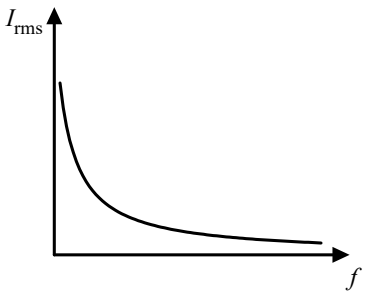
$$\frac{dI}{dt} = \frac{2.50}{0.350} = 7.14 \text{ A s}^{-1}; L = \frac{\varepsilon}{dI/dt} = 2.10 \times 10^{-3} \text{ H}$$

$$A = \frac{Ll}{\mu_0 N^2} = \frac{(2.10 \times 10^{-3})(0.0800)}{(4\pi \times 10^{-7})(200)^2} = 3.34 \times 10^{-3} \text{ m}^2$$

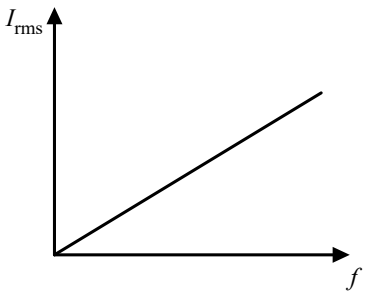
Explanation: Combine $\varepsilon = L\, dI/dt$ with $L = \mu_0 N^2 A/l$.

14. An alternating current power supply with a constant root mean square voltage and variable frequency is connected in series to a pure inductor. Which graph shows the variation of the root mean square current, I_{rms}, with the frequency, f?

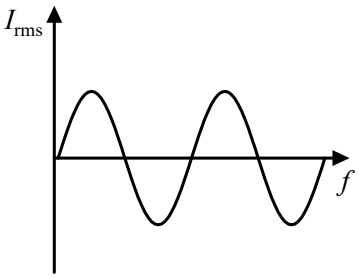
A



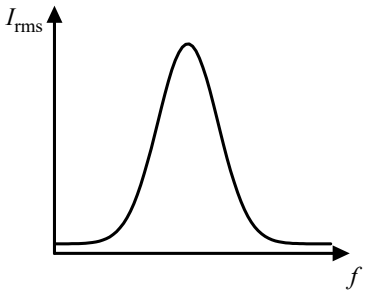
B



C



D

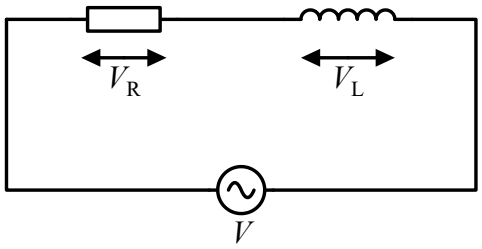


Answer: A

$$I_{rms} = \frac{V_{rms}}{2\pi fL} \propto \frac{1}{f} \text{ (A)}$$

Explanation: Inductive reactance increases with frequency.

15. A pure inductor and a pure resistor are connected in series with an alternating voltage, V, as shown in the circuit diagram below.



The potential differences across the resistor and inductor are V_R and V_L respectively. Which expression shows the correct relation between V, V_R and V_L?

A $V = \sqrt{(V_R^2 + V_L^2)}$ B $V = \sqrt{(V_R^2 - V_L^2)}$ C $V = V_R + V_L$ D $V = V_R - V_L$

Answer: A

V_R and V_L are 90° out of phase, so they add as vectors: $V = \sqrt{(V_R^2 + V_L^2)}$.

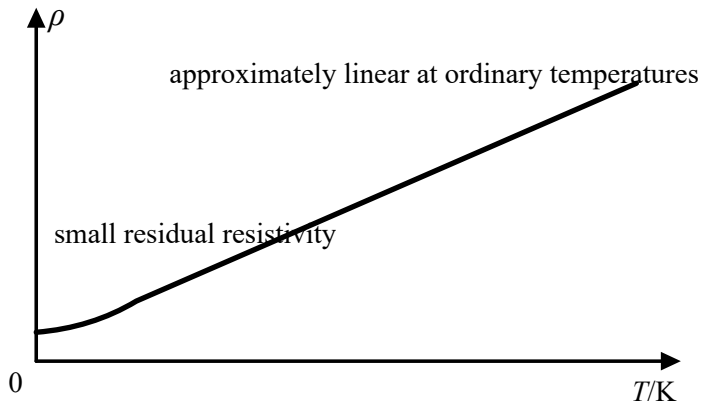
Explanation: Simply adding the r.m.s. values would ignore the phase difference.

Section B [15 marks]

16. A transmission power line that is made of copper is exposed to hot and cold weathers.

(a) Sketch a graph to show the effect of temperature change, in Kelvin, on the resistivity of the copper. [3]

Answer



Axes: resistivity against temperature (K)	1
Resistivity increases with temperature (approximately linearly)	1
Small non-zero resistivity near 0 K	1
	3

Explanation: As temperature rises the lattice vibrates more, electrons collide more often and resistivity increases.

(b) Besides temperature, state **two** factors that affects the resistance of the transmission power line. [2]

Answer

Length of the wire	1
Cross-sectional area (thickness) of the wire / type of material	1
	2

Explanation: $R = \rho l/A$.

(c) If the copper wire has the resistivity of 1.72×10^{-8} Ω m, length 5.0 km and resistance 1.0 Ω, calculate the diameter of the wire of the transmission power line. [3]

Answer

$R = \frac{\rho L}{A}, A = \frac{\pi d^2}{4} \Longrightarrow d = \sqrt{(\frac{4\rho L}{\pi R})}$ 1

$d = \sqrt{(\frac{4(1.72 \times 10^{-8})(5.0 \times 10^{3})}{\pi(1.0)})}$ 1

$d = 1.05 \times 10^{-2}$ m 1

3

Total for question **8**

Explanation: Convert 5.0 km to 5.0×10^3 m.

17. An oven is labelled 240 V, 50 Hz and 1200 W. The oven is connected to a power supply of 240 V.

(a) Calculate the root mean square current that is supplied to the oven. [2]

Answer

$I_{rms} = \frac{P}{V_{rms}} = \frac{1200}{240}$ 1

= 5.0 A 1

2

Explanation: Mains voltages are quoted as r.m.s. values.

(b) Calculate the peak current that is supplied to the oven. [2]

Answer

$I_0 = \sqrt{2}\, I_{rms} = \sqrt{2}(5.0)$ 1

= 7.07 A 1

2

Explanation: Peak = √2 × r.m.s. for a sinusoidal current.

(c) Between 10 A and 13 A fuses, which is more suitable to be used to supply current to the oven? Give your reasons. [3]

Answer

The 10 A fuse	1
Its rating is just above the operating current (5.0 A r.m.s., 7.07 A peak), so it does not blow in normal use	1
but it melts sooner than a 13 A fuse if a fault causes a large current, giving better protection	1
	3
Total for question	**7**

Explanation: A fuse should be rated slightly above the normal operating current.

Section C [30 marks]

18. (a) A point charge, $+Q$, is in an isolated region as shown in the diagram below.



Describe the electric force experienced by an electron placed at P in the region. [3]

Answer

The electron is attracted towards $+Q$ (force directed from P towards $+Q$)	1
Magnitude $F = \frac{Qe}{4\pi\varepsilon_0 r^2}$	1
The force increases as the electron gets closer (inverse square law)	1
	3

Explanation: Unlike charges attract; the force acts along the line joining the charges.

(b) Describe the motion of an electron that enters perpendicularly to a uniform electric field. [4]

Answer

The electron experiences a constant force eE opposite to the field (towards the positive plate)	1
It has a constant acceleration perpendicular to its initial velocity	1
Its velocity parallel to the plates stays constant	1
So it follows a parabolic path (like a horizontally projected object)	1
	4

Explanation: The motion is analysed like projectile motion.

(c) An electron experienced a force of 2.0×10^{-15} N enters at a middle of parallel charged plates with velocity of 1.7×10^{7} m s^{-1}. If the length of the plates is 8.0 cm and the separation between the plates is 4.0 cm, determine whether the electron could exit or not from the other end of the plates. [8]

Answer

$a = \frac{F}{m} = \frac{2.0 \times 10^{-15}}{9.11 \times 10^{-31}}$	1
$a = 2.20 \times 10^{15}$ m s^{-2}	1
The electron enters midway, so it hits a plate if it is deflected by 2.0 cm	1
Time to be deflected 2.0 cm: $0.020 = \frac{1}{2}at^2$	1
$t = 4.27 \times 10^{-9}$ s	1
Horizontal distance in this time = $vt = (1.7 \times 10^{7})(4.27 \times 10^{-9})$	1
$= 0.073$ m $= 7.3$ cm	1
7.3 cm < 8.0 cm, so the electron hits the plate and cannot exit	1
	8
Total for question	**15**

Explanation: Equivalently, over the full 8.0 cm the deflection would be 2.43 cm, more than the 2.0 cm available.

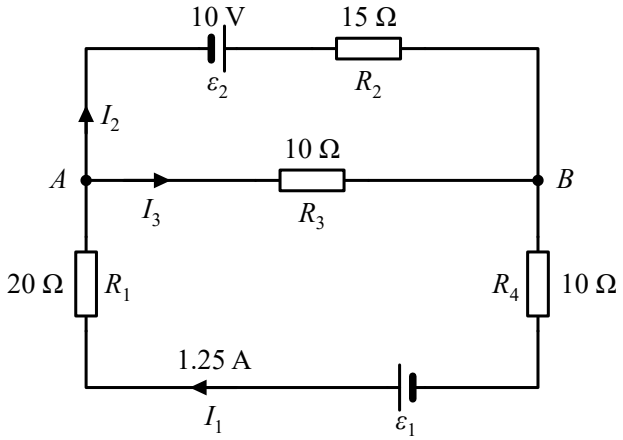
19. (a) State **two** principles applied in Kirchhoff's law. [2]

Answer

Conservation of charge (junction rule): total current into a junction = total current out	1
Conservation of energy (loop rule): in any closed loop, sum of e.m.f.s = sum of p.d.s	1
	2

Explanation: These are Kirchhoff's first and second laws.

(b) A circuit consists of two batteries, ε_1 and ε_2, with a negligible internal resistance connected to four resistors, R_1, R_2, R_3 and R_4, is shown in the diagram below.



I_1, I_2 and I_3 are the current flow in the circuit.

(i) Calculate ε_1 if the potential difference of A to B is 3.5 V. [3]

Answer

Loop containing ε_1, R_1, $A \rightarrow B$ and R_4: $\varepsilon_1 = I_1(R_1 + R_4) + V_{AB}$ — 1

$\varepsilon_1 = (1.25)(20 + 10) + 3.5$ — 1

$\varepsilon_1 = 41$ V — 1

3

Explanation: The p.d. from A to B is the same whichever branch is used.

(ii) Determine I_2 and I_3. [4]

Answer

Middle branch: $I_3 = \frac{V_{AB}}{R_3} = \frac{3.5}{10}$ — 1

$I_3 = 0.35$ A — 1

Junction A: $I_2 = I_1 - I_3 = 1.25 - 0.35$ — 1

$I_2 = 0.90$ A (check: top branch $0.90 \times 15 - 10 = 3.5$ V) — 1

4

Explanation: The check through the top branch confirms the answer.

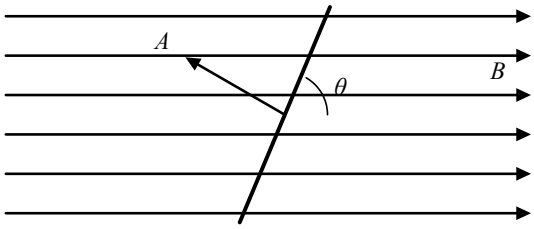
(c) Explain the use of shunts and multipliers in galvanometer. [6]

Answer

Shunt: a low resistance connected in parallel with the galvanometer	1
Most of the current bypasses the galvanometer, converting it into an ammeter for large currents	1
$R_S = \frac{I_g R_g}{I - I_g}$	1
Multiplier: a high resistance connected in series with the galvanometer	1
It limits the current so that a large p.d. gives full-scale deflection, converting it into a voltmeter	1
$R_m = \frac{V}{I_g} - R_g$	1
	6
Total for question	**15**

Explanation: An ammeter needs low resistance; a voltmeter needs high resistance.

20. (a) A wire loop that has an area A at an angle θ in a uniform magnetic field B is shown in the diagram below.



(i) Write an expression of magnetic flux through the loop. [1]

Answer

$\Phi = BA \cos\theta$, where θ is the angle between **B** and the normal to the loop	1
	1

Explanation: Flux is greatest when the field is perpendicular to the plane of the loop.

(ii) State **three** ways that can induce e.m.f. in the loop. [3]

Answer

• Change the magnetic flux density B	1
• Change the area A of the loop inside the field	1
• Change the angle θ (rotate the loop)	1
	3

Explanation: Any change in flux linkage induces an e.m.f.

(b) A solenoid has a length l, cross sectional area A and number of turns N. Derive the self-inductance of the solenoid. [5]

Answer

Field inside: $B = \frac{\mu_0 NI}{l}$ 1

Flux through each turn: $\Phi = BA = \frac{\mu_0 NIA}{l}$ 1

Flux linkage $N\Phi = \frac{\mu_0 N^2 IA}{l}$ 1

$N\Phi = LI$ 1

$L = \frac{\mu_0 N^2 A}{l}$ 1

5

Explanation: Self-inductance is flux linkage per unit current.

(c) An a.c. generator consists of a coil of N turns with an area 2.0×10^{-2} m^2. The coil rotates in a uniform magnetic field of 0.15 T at 50.0 Hz. If the root mean square voltage produced by the generator is 240 V, determine the number of N. [6]

Answer

Peak e.m.f. $\varepsilon_0 = NBA\omega$ 1

$\omega = 2\pi f = 2\pi(50.0) = 314$ rad s^{-1} 1

$\varepsilon_0 = \sqrt{2} \times 240 = 339$ V 1

$N = \frac{\varepsilon_0}{BA\omega}$ 1

$= \frac{339}{(0.15)(2.0 \times 10^{-2})(314)}$ 1

$N = 360$ turns 1

6

Total for question **15**

Explanation: The given 240 V is r.m.s.; the generator formula gives the peak e.m.f.